Varifa.

№	y-actual	y-pred
1	2	1.5
2	4	3
3	4.5	4
4	3	2
5	5	5.5
6	6	6.5
7	5.5	6.5
8	7	8
9	4	3
10	8	~~8~~ 9

$$MSE = \frac{\sum_{i=1}^{n} (y_i - y_p)^2}{n}$$

$$MSE = \frac{0.5^2 + 1^2 + 0.5^2 + (1)^2 + (0.5)^2 + (0.5)^2 + (-0.5)^2 + (-1)^2 + 1^2 + (-1)^2}{10} =$$

$$\cancel{= \left(\frac{6 \cdot 0.25 + 4}{10}\right) = \frac{5.5}{10} = 5.5}$$

$$= \frac{4 \cdot 0.25 + 6}{10} = \frac{1+6}{10} = 0.7$$

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{0.7} \approx 0.84$$

$$MAE = \frac{\sum_{i=1}^{n} |y_{act} - y_{pred}|}{n}$$

$$MAE = \frac{4 \cdot 0.5 + 6}{10} = \frac{8}{10} = 0.8$$

Caption

$$R2 = 1 - \frac{MSE \cdot n}{\text{Total error}}$$

$$R2 = 1 - \frac{MSE \cdot n}{\Sigma (y_i - \bar{y}_i)^2}$$

$\bar{y}_i$ = { Mean of Actual

$$\bar{y}_i = \frac{2+4+4.5+3+5+6+5.5+7+4+8}{10} = \frac{49}{10} = 4.$$

$$\Sigma (y_i - \bar{y}_i)^2 = (2-4.9)^2 + (4-4.9)^2 + (4.5-4.9)^2 \ldots + (8-4.9)^2 = 8.41 + 0.81 + 0.16 + 3.61 + 0.01 + 1.21$$
$$4.41 + 0.81 + 9.61 = 29.4.$$

$$\Sigma (y_i - \bar{y}_i)^2 = 29.4.$$

$$R2 = 1 - \frac{7}{29.4} = 1 - 0.24 = 0.76$$

Caption